

$$\textcircled{1} \quad f(x) = x - \frac{2}{x} \quad X = [1, 2]$$

Newton - Verfahren : konvergiert
(Kontraktion, α)

$$\phi(x) = x - \frac{f(x)}{f'(x)} \quad , \quad f'(x) = 1 + \frac{2}{x^2}$$

$$= x - \frac{x - \frac{2}{x}}{1 + \frac{2}{x^2}} \cdot \frac{x^2}{x^2}$$

$$= x - \frac{x^3 - 2x}{x^2 + 2}$$

$$= \frac{x(x^2 + 2) - (x^3 - 2x)}{x^2 + 2}$$

$$= \frac{x^3 + 2x - x^3 + 2x}{2 + x^2}$$

$$\Rightarrow \phi(x) = 4 \frac{x}{2 + x^2}$$

BFS : ϕ muss Kontraktion auf $X = [1, 2]$

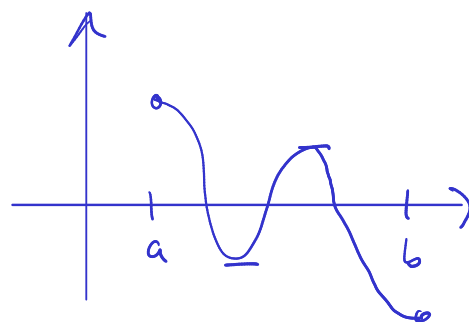
$$(i) \quad \phi(x) \in X \quad \forall x \in X$$

$$(ii) \quad \exists \alpha < 1 \quad \text{mit}$$

$$|\phi(y) - \phi(x)| \leq \alpha |y - x| \quad \forall x, y \in X$$

$$(i) \quad \min_{x \in X} \phi(x) \quad , \quad \max_{x \in X} \phi(x)$$

$\uparrow$ $\uparrow$
 X X



lokale Extrema von ϕ : $\phi'(x) = 0$

$$\phi(x) = 4 \frac{x}{2+x^2}$$

$$\phi'(x) = 4 \frac{1 \cdot (2+x^2) - x \cdot 2x}{(2+x^2)^2}$$

$$= 4 \frac{2+x^2-2x^2}{(2+x^2)^2}$$

$$= 4 \frac{2-x^2}{(2+x^2)^2}$$

$$\phi'(x) = 0 \Leftrightarrow x^2 = 2 \Leftrightarrow x = \pm \sqrt{2}$$

$$X = [1, 2] \Rightarrow x = \sqrt{2}$$

$$\phi(\sqrt{2}) = \sqrt{2} \in X$$

Rand: $\phi(1) = \frac{4}{3}$
 $\phi(2) = \frac{4}{3}$ $\searrow \in [1, 2] = X$

(ii) Kontraktionszahl α

MWS : $\alpha \leq \max_{\xi \in X} |\phi'(\xi)|$

$$\phi'(x) = 4 \frac{2-x^2}{(2+x^2)^2}$$

$$\phi''(x) = \frac{4 \cdot (-2x)}{(2+x^2)^3} = 8 \frac{x(x^2-6)}{(2+x^2)^3} \quad |$$

lokale Extrema von ϕ' : $\phi''(x) = 0$

$$\Leftrightarrow x = 0 \vee x^2 = 6$$

$$\Rightarrow \text{alle } x \notin X$$

Ränder : $\phi'(1) = \frac{4}{9}$

$$\phi'(2) = -\frac{2}{9}$$

$$\alpha \leq \max\left(\left|\frac{4}{9}\right|, \left|-\frac{2}{9}\right|\right) = \frac{4}{9} < 1$$

$\Rightarrow$ BFS "geht"

$\Rightarrow$ Newton konv. $\forall x_0 \in X = [1, 2]$